

Impact of 3D Integration on Thermal Performance of RISC-V MemPool Multi-Core SOC

Sankatali Venkateswarlu, Subrat Mishra, Herman Oprins, *Member, IEEE*, Bjorn Vermeersch, Moritz Brunion, Jun-Han Han, *Member, IEEE* Mircea R. Stan, *Fellow, IEEE*, Dwaipayan Biswas, Pieter Weckx, and Francky Catthoor, *Fellow, IEEE*

Abstract— Due to the rise in the number of cores in modern multi-core architectures, 3D integration (i.e., vertical stacking of chips) of System-on-a-chip (SOC) promises better performance due to a drastic reduction in global interconnect lengths and die footprint compared to 2D counterparts. However, thermal issues are predominant in 3D-SOCs due to the vertical stacking nature of chips which multiplies the transistor power density by the number of dies within the stack. Also, the reduced lateral heat spreading with aggressive die thinning, degrades the on-chip thermal performances. In this paper, we investigate the thermal performance analysis of 3D-SOC and compare the results with the 2D-SOC designs for a MemPool multi-core SOC with shared L1 scratchpad memory (SPM). Simulation results reveal that the 3D-SOC using Memory on Logic configuration increases the on-chip maximum temperature by more than 20% compared to the baseline 2D-SOC and the logic die temperature is relatively higher (3.6%) than the memory die. We also explore the impact of architectural floor-planning effects and 3D functional partitioning on thermal performance of the MemPool instances in the 3D-SOC with memory capacity ranging from 1 MiB to 8 MiB and benchmarked the thermal performance with the 2D-SOC designs. We observe that the junction to ambient temperature (T_{max}) increases by 44% and is predominant for the scratchpad memory (SPM) capacity of 8MiB. Further investigations on various 3D stacking configurations reveal there is an improvement in thermal performance for memory-on-logic (MOL) over logic-on-memory (LOM) for L1 SPM capacity of 1 MiB, 2 MiB and 4 MiB, and LOM over the MLL configuration for L1 SPM capacity of 8 MiB.

Index Terms—MemPool 3D, system-on-a-chip (SOC), multi-core architecture, Memory-on-Logic (MOL), Logic-on-Memory (LOM), workload, on-chip temperature, lateral thermal spreading.

I. INTRODUCTION

OVER the past few decades, the continued technology scaling (following Moore's law) of microelectronic devices with advancement of process technology has led to a remarkable improvement in performance of electronic products. However, with further scaling, the device performance

gets degrade due to intrinsic physical limitations such as short channel effects, increased contact resistances and degraded device reliability (HCD [1], TDDDB [2], BTI [3], self-heating effects in advanced 3D transistors [4], [5] and leads to more and more challenging from both technological and financial perspective. In order to continue the performance gain, semiconductor industry moved to system technology scaling rather device scaling by incorporating multi-core architectures. Therefore, the evolution of multi-core architectures leads the semiconductor industry to incorporate more and more number cores in Systems-on-a-Chip (SOC) to increase the performance. According to the literature, the core count in a silicon chip doubles with every new technology generation and soon, a single chip can hold up to thousand cores [6], [7]. However, with every new technology node the combined effect of the increase in core count scaling and the relaxation of transistor scaling results an increase in die area footprint [7]. Furthermore, the rise in footprint of the 2D-SOC is severely limited by routing congestion and longer interconnect delays. In order to tackle these problems, the industry is moving to 3D-SOC technology, where the SOC is functionally partitioned (e.g., memory, logic, I/O) across layers which are stacked vertically [8], [9]. This reduces not only the die area footprint but also the interconnect routing length [10] and hence further improvement in Power-Performance-Area (PPA) metric due to a reduction in interconnect wiring resistance and capacitance [11]. Also, a number of innovative process technologies go into the realization of a working 3D-SOC such as wafer thinning, etching and filling of high aspect ratio holes in silicon, inter-strata bonding etc. [12], [13]. In addition, a variety of integration technologies for 3D integration such as face-to-face integration [12] and back-to-face integration [14] have been demonstrated.

However, with stacking chips vertically, although the power consumption of 3D-SOCs is expected to decrease (due to the shorter interconnect lengths), the power density increases greatly compared to 2D-SOC designs. This is due to the presence of larger number of transistors per unit area (depending on the number of dies in the stack) in the 3D-SOC. Therefore, thermal issues are expected to become more dominant in 3D-SOCs compared to conventional 2D systems, which necessitates expensive cooling system and is a major problem [15], [16], [17] [18]. There are some dynamic thermal management (DTM) techniques include dynamic voltage and frequency scaling (DVFS), task migration, fetch gating, decode throttling, speculation control, stop & go, and clock throttling

Manuscript received XXXX XX XXXX; revised XXXX XX XXXX; accepted XXXX XX XXXX. Date of publication XXXX XX XXXX; date of current version XXXX XX XXXX. (Corresponding author: Sankatali Venkateswarlu.)

Sankatali Venkateswarlu and Francky Catthoor are with IMEC, 3001 Leuven, Belgium, and also with the Department of Electrical Engineering (ESAT), Katholieke Universiteit Leuven (KU Leuven), 3000 Leuven, Belgium.

Subrat Mishra, Herman Oprins, Bjorn Vermeersch, Moritz Brunion, Dwaipayan Biswas and Pieter Weckx are with IMEC, 3001 Leuven, Belgium.

Jun-Han Han and Mircea R. Stan are with the Department of Electrical and Computer Engineering, University of Virginia, Charlottesville, VA 22904 USA.

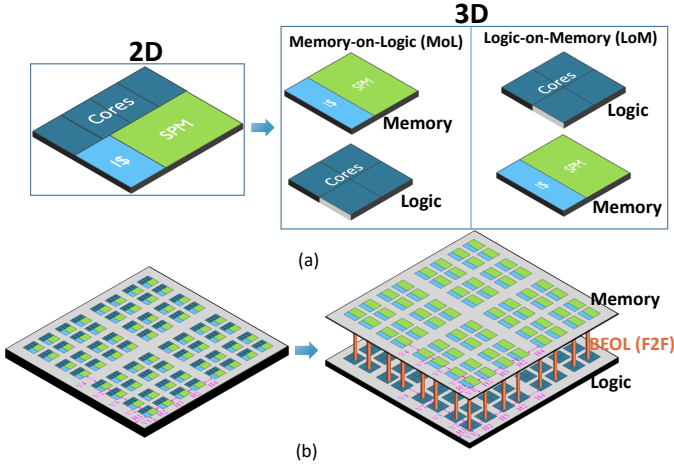


Figure 1: SOC functional partitioning from 2D to 3D (a) tile level and (b) cluster level implementation.

[19], [20], [21]. However, the DTM cannot be optimized in multicore 3D-SOCs solely based on these techniques because it does not consider the vertical heat spreading, which is more significant in 3D-SOCs as compared to 2D-SOCs [22], [23]. In order to tackle this issue few research groups developed a DTM technique applied to the thermal-aware 3D processor architectures [24], [25] [26]. Furthermore, the silicon layer needs to be thinned down to a few microns in order to enable nano-Through Silicon-Vias (nTSV) to communicate between bottom dies (i.e., I/O layer) and top die (i.e., logic/memory layers) up to the package system [27], [28], [29]. But the silicon logic layer thinning degrades the lateral heat spreading due to decrease in thermal conductivity (k_{th}) of thinned silicon and hence leads to a significant degradation of the SOC thermal performance. Also, the on-chip and inter die thermal coupling (under heavy workload conditions) arises and increases with the rising number of cores per a chip in 3D-SOCs. Therefore, there is a significant demand for system-level thermal modeling and optimization in order to improve the thermal performance of multi-core 3D-SOCs. There are some literature reports on thermal analysis in multi-core 3D integrated circuits with analytical and finite-element models of heat transfer [30]. The thermal implications considering the impact of 3D IC integration designs such as Face-to-Face (F2F) and Face-to-Back (F2B) are reported in [31], the number of stacking layers [32], Micro-bump, TSV and TMV [33] on thermal performance of multi-core 3D-SOC architectures. Also, there are few literature reports investigated the multi-layer cooling system to reduce thermal hotspots in 3D ICs [34]. Additionally, the workload activities will also strongly impact the thermal performance of the SOC [35], [36], [37]. Moreover, the core activity patterns significantly affect the thermal performance of the multi-core SOC under partial workload conditions as reported in our previous work [38]. Additionally, thermal profiles are critical for on-chip thermal sensor placement, T_{max} control, and thermal-aware EM sign-off [39].

In this paper we investigate the aforementioned effects on thermal performance of the MemPool 3D-SOC and compare

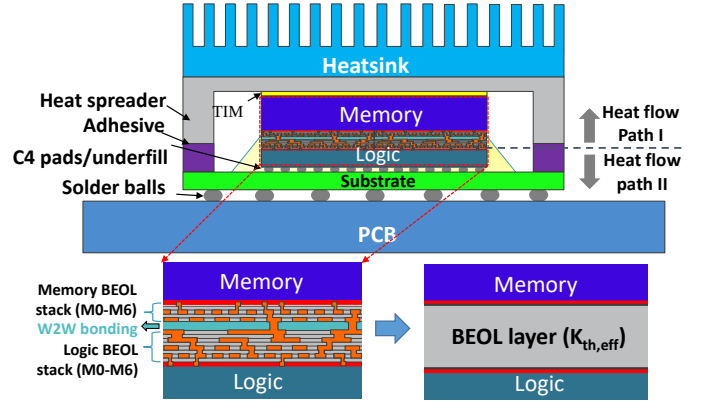


Figure 2: Schematic representation of the MemPool 3D-SOC (using memory on logic) with package information. Here, the BEOL stack (M6M6, 12 layers) along with F2F via is assumed with a single stack of layer with effective thermal conductivity.

the results with the 2D counterparts. Following the introduction, in section II, we describe the full 3D-SOC design considerations along with package geometry, physical parameters and the simulation methodology for thermal modeling. Simulation results for thermal performance comparison between 2D and 3D-SOC are discussed in section III. Also the impact of Si logic layer thinning, and the results of applying well-chosen synthetic tile activity patterns under partial workload conditions are discussed in this section. In section IV, we discuss the impact of scratchpad memory (SPM) capacity on thermal performance of MemPool 3D-SOC. The impact of 3D stacking configuration such as MOL and LOM are discussed on section V. Finally, we draw conclusions in section VI.

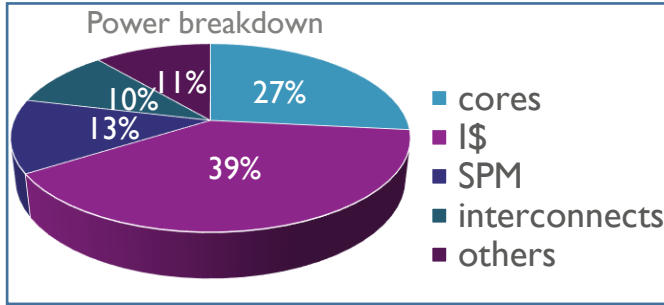
II. SOC CONFIGURATION WITH PACKAGE PHYSICAL AND THERMAL DESIGN PARAMETERS

In this paper, we consider MemPool, a 32-bit RISC-V based multi-core (256-core) architecture with low-latency shared-L1 memory [40] as a test case for thermal performance comparison. In the MemPool architecture there are 64 tiles with each tile containing 4 cores sharing a large pool of 16 KiB scratchpad memory (SPM) blocks along with 2KiB of instruction cache (I\$) blocks. In MemPool 3D the complex SOC is functionally partitioned into a stack of the memory layer (with I\$ and SPM) and the logic layer (with snitch cores) [41]. With this functional partitioning we can perform thermal modeling and benchmark the thermal performance between 2D and 3D-SOCs. The schematic of the design in tile and cluster level implementation for the proposed MemPool SOC architecture from 2D to 3D are shown in Fig. 1.

The schematic of the MemPool 3D-SOC using memory on logic configuration with full package structure is shown in Fig. 2. In the 3D implementation, we consider face-to-face (F2F) wafer-to-wafer hybrid bonding between two dies (i.e., logic die and memory die). This will enable in lowering of interconnect pitches and hence higher interconnect density for maintaining the PPA gain. The 3D back end of line (BEOL) consists of six metal layers in both tiers (M6M6), separated by the F2F via layer of thickness $2.85 \mu\text{m}$ [42]. The baseline MemPool

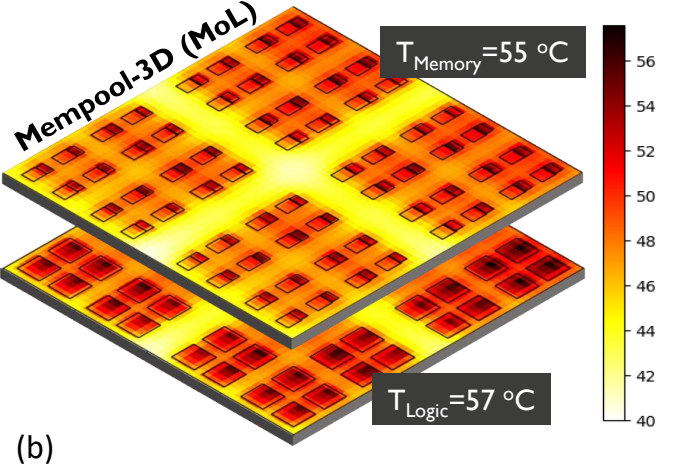
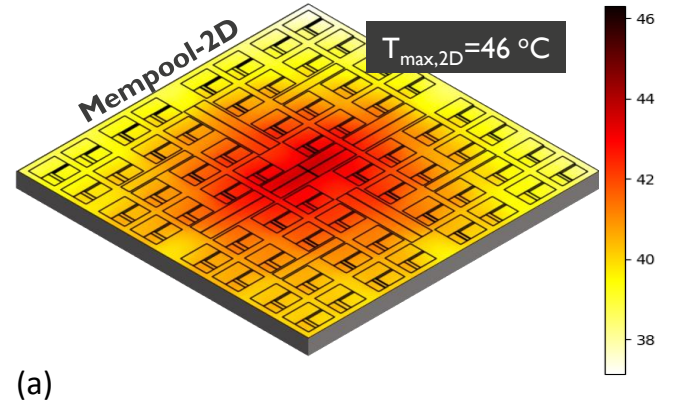
Table I: MemPool 3D-SOC physical and thermal parameters along with full Package design considered in thermal modeling.

Layer name	Size (mm^2)	thickness (μm)	Thermal conductivity ($W m^{-1} K^{-1}$)
Memory die	3.95×3.95	100	140
BEOL stack	3.95×3.95	2.85	1.2
Logic die	3.95×3.95	5	100
TIM	3.95×3.95	100	7
Heat spreader	5.92×5.92	200	400
Heatsink	7.9×7.9	1000	400
C4 micro bumps	3.95×3.95	10	1.25
Package substrate	10×10	100	2
Solder balls	10×10	940	16.67
Under fill	10×10	940	33.33
PCB	15×15	2000	3

**Figure 3:** Breakdown of the thermal design power (TDP) within the MemPool tile.

2D-SOC footprint is $4.8 \times 4.8 \text{ mm}^2$ with each tile size of $425 \times 425 \mu m^2$ on the silicon die. The utilization of tiles within the die is 50%. The detailed floorplan information of the Memopool 2D-SOC architecture is given in [38]. However, in MemPool 3D-SOC, the tile is functionally partitioned into two blocks: memory block and logic block. The partitioning scheme helps to reduce the size of each tile by a factor of 0.667, and this helps scale down both memory and logic die footprint by a factor of 0.665 for the L1 SPM memory capacity of 1MiB [41]. The 3D-SOC is mounted on a PCB with a layer of solder balls and a package substrate with C4 micro-bumps using flip-chip packaging technology.

In our thermal modeling study, we have replaced the C4 and solder balls with a single stack of layer with their equivalent thermal conductivities ($k_{th,eff}$) of 1.25 and 16.67 $W m^{-1} K^{-1}$, respectively. The thermal conductivity for silicon in the memory die and logic die are 140 and 100 $W m^{-1} K^{-1}$ for their thicknesses of 100 and 5 μm respectively [43]. The lower k_{th} for the logic die compared to the memory die is attributed to the thinning of the silicon logic layer from 100 to 5 μm . The thermal interface material (TIM) between the silicon die and heatspreader plays an important role in heat conduction from the die to the heatsink. There are various types of TIM materials (e.g., polymer, film, metal etc.) based the package thermal solution [44]. In our study we have considered 100 μm thickness of polymer type TIM material (PTIM) with k_{th} of 7 $W m^{-1} K^{-1}$. The heat spreader and heatsink base are of thickness 0.2 mm and 1 mm, respectively, with k_{th} of 400 $W m^{-1} K^{-1}$. On top of the heatsink, we assume a cooling solution with a convective resistance (R_{conv})

**Figure 4:** Thermal profile of the proposed SOC architectures. (a) MemPool-2D SOC and (b) MemPool-3D SOC (using memory on logic).

of 0.17 K/W between the heatsink and the ambient. On the other side towards PCB, we assume natural convection with R_{conv} of 50 K/W between printed circuit board (PCB) and the ambient. The detailed physical and thermal modeling parameters of the MemPool 3D-SOC are given in Table I.

In this work, we assume that the proposed MemPool SOC is designed in a commercial 28nm technology, which is a representative node for the target domain of this RISC-V core [40]. We also assume that the thermal design power (TDP) of the 2D-SOC is 40 W (for larger workloads) as a base line for thermal performance comparative study [38]. The detailed power breakdown within the tiles and interconnect routing blocks (both local and global) according to the reference [40] are depicted in Fig. 3. However, in MemPool 3D-SOC, the smaller die area footprint and the reduction in interconnect wiring length reduces the number of buffers, which lowers the power dissipation (0.913 x). However, the power density in the MemPool-3D rises sharply (x 1.37) over the 2D-SOC due to the overall reduction in area footprint for the 3D-SOC.

III. SOC THERMAL PERFORMANCE ANALYSIS: 2D vs 3D

In this work, we have used HOTSPOT v7.0 [45],[46] to perform steady state thermal modeling. The HotSpot thermal tool is validated against commercial tool ANSYS [47] which is a detailed finite element simulator for 3D heat flow. The

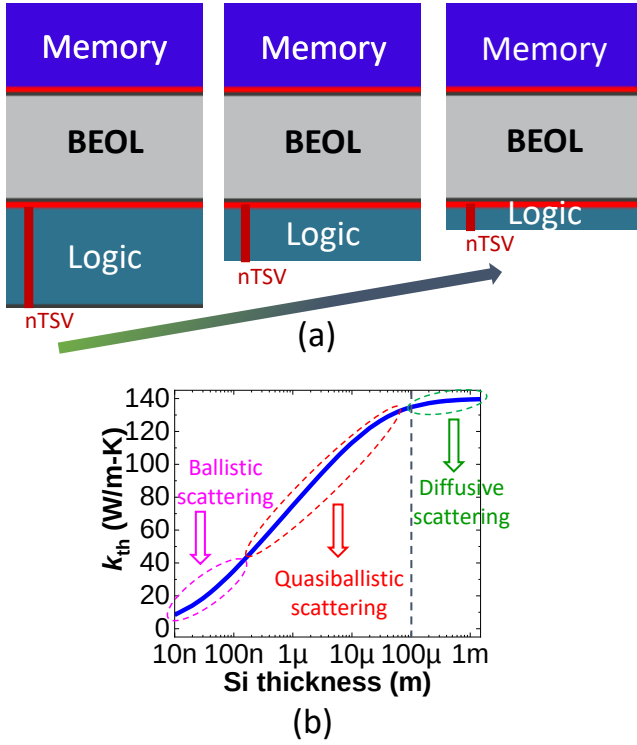


Figure 5: (a) Silicon logic layer thinning to enable nano through silicon vias (nTSVs). (b) Thermal conductivity (k_{th}) versus silicon layer thickness. The k_{th} decreases faster rate below 100 μm .

HotSpot modeling method is also validated by comparing with a commercial thermal testing chip [48]. The tool has the capability for high-level floorplanning with user defined area. It also has the ability to model dynamic localized heating in and among the architectural-level functional blocks by constructing an equivalent thermal RC network model based on lumped thermal resistances and capacitances in both block and grid level and solves using a finite difference method. However, the tool has some limitations that it assumes uniform power density within a block intended to use early pipeline-level microarchitecture floorplanning and optimization. Here, we assume that the ambient temperature is 27°C, which is a representative value for such studies and the temperature values reported in this paper are the values above the ambient.

A. 2D-SOC vs 3D-SOC

Fig. 4 shows the thermal profile of the MemPool 2D-SOC and 3D-SOC with the logic and memory layer temperature distributions above the ambient. From the figure, it can be observed that the maximum temperature (T_{max}) with respect to ambient is 46°C for 2D-SOC, whereas for 3D-SOC the T_{max} is 55°C in the memory die and 57°C in the logic die. Thus, for 3D-SOC in both the logic and memory die, the T_{max} is increased by 24% and 20% respectively. It can also be observed that the hotspot temperature in the logic die is relatively (3 °C) higher than in memory die. This is due to the increased distance of the logic layer from the cooling solution (i.e., heatsink), which increases the thermal resistance of the logic layer towards the heatsink due to the presence of the additional low thermal conductivity BEOL stack layer.

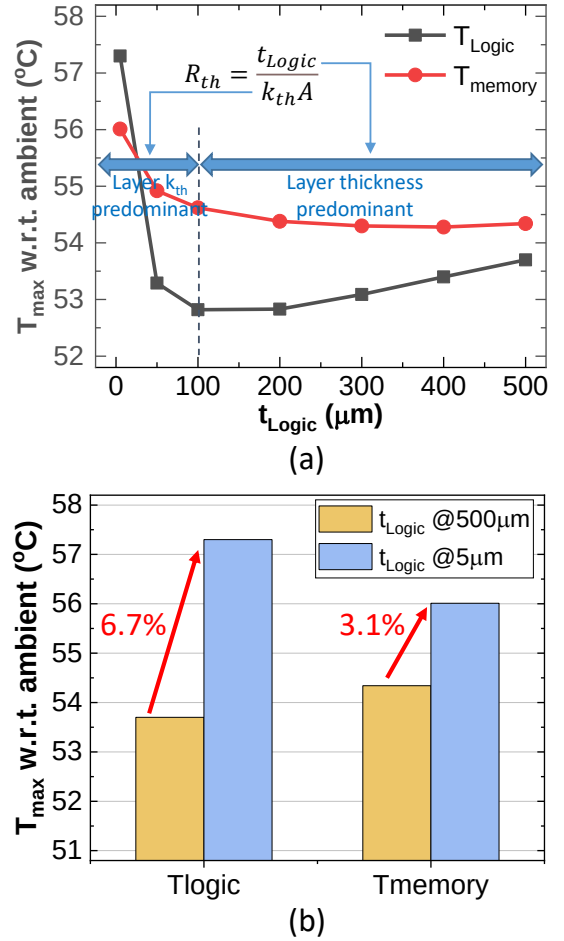


Figure 6: (a) The rise in maximum on-chip temperature (T_{max}) of the logic and memory layer versus Si logic layer thickness and (b) the % rise in T_{max} for silicon logic die thickness of 5 μm compared 500 μm .

B. Impact of Si Logic layer thinning on SOC Thermal Performance

In order to enable nano-Through-Silicon-Vias (nTSV) between two Si dies for communicating between logic/memory and input/output (IO) layer and, to the package substrate[33], the silicon logic layer needs to be thinned as shown in Fig. 5(a). However, the Silicon thermal conductivity decreases with decreasing the layer thickness which is shown in Fig. 5(b)). Moreover, for below 100 μm , the k_{th} decreases with faster rate as highlighted in the figure due to quasiballistic and ballistic phonon scattering effect [43]. In this section, we investigate the impact of silicon logic layer thinning on MemPool 3D-SOC on-chip temperature and lateral thermal spreading. From Fig. 6(a), with decreasing the silicon substrate thickness of the logic die, the temperature of the die is decreased up to a thickness of 100 μm due to decrease in distance from the logic layer to the PCB. This also enables the memory layer to enhance the heat conduction towards the PCB and so the memory temperature also decreases as shown in the figure. However, with further thinning the Si logic layer below 100 μm , the thermal conductivity of the logic layer decreases at a faster rate than the decrease in thickness. This ultimately

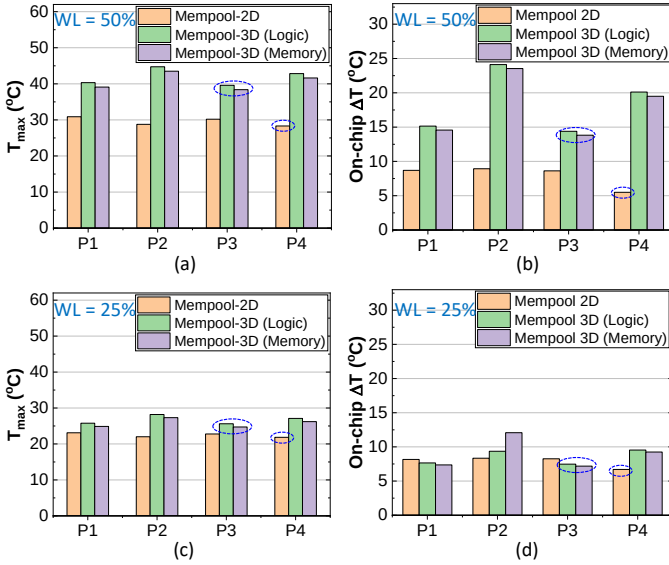


Figure 7: (a) On-chip maximum temperature (T_{max}) and (b) thermal gradient (ΔT) comparison between 2D vs 3D SOC for different tiles activity patterns under 50% workload. (c) T_{max} (d) ΔT with 25% workload.

increases the Si layer thermal resistance towards the PCB. Also, the lateral thermal resistance increases at an even higher rate and therefore the temperature rises sharply for a silicon thickness below $100 \mu m$. The sharp rise in temperature in the logic layer further increases the heat energy coupling from the logic die towards the memory die and hence the memory layer temperature also increases. Therefore, the silicon logic layer thinning significantly impacts the thermal performance of the 3D-SOC by rising the logic and memory layer temperature by 6.7% and 3.1% respectively as shown in Fig. 6(b). The simulation results are in agreement with the literature [49].

C. Tiles Activity Pattern on SOC thermal Performance: 2D vs 3D

It is highly desirable to obtain the optimum core-level activity patterns in the multi-core systems in order to improve the power and thermal performance under real workload conditions. For example, a core level several well-chosen synthetic activity prediction patterns are proposed in [50], [51]. Therefore, under partial workload conditions, assigning workload to a specific core/tile in terms of both power and thermal point of view without degrading the system performance is an important task. In this section, we investigate and compare the thermal performance of MemPool 2D-SOC and 3D-SOC by considering the impact of different tiles activity patterns under partial workload. The case study is performed under 50% workload with four different tile activity patterns (P1, P2, P3 and P4) considered in [38]. Here, 50% workload means, only 50% of tiles in the SOC (i.e., 32 tiles) are active and remaining are idle state. Fig. 7 and Fig. 8 show the thermal performance comparison results and the on-chip temperature map for the proposed four tile activity patterns of the 2D and 3D-SOCs. It is clearly observed that for all the tile activity patterns, the thermal performance is poor for the 3D-SOC compared the

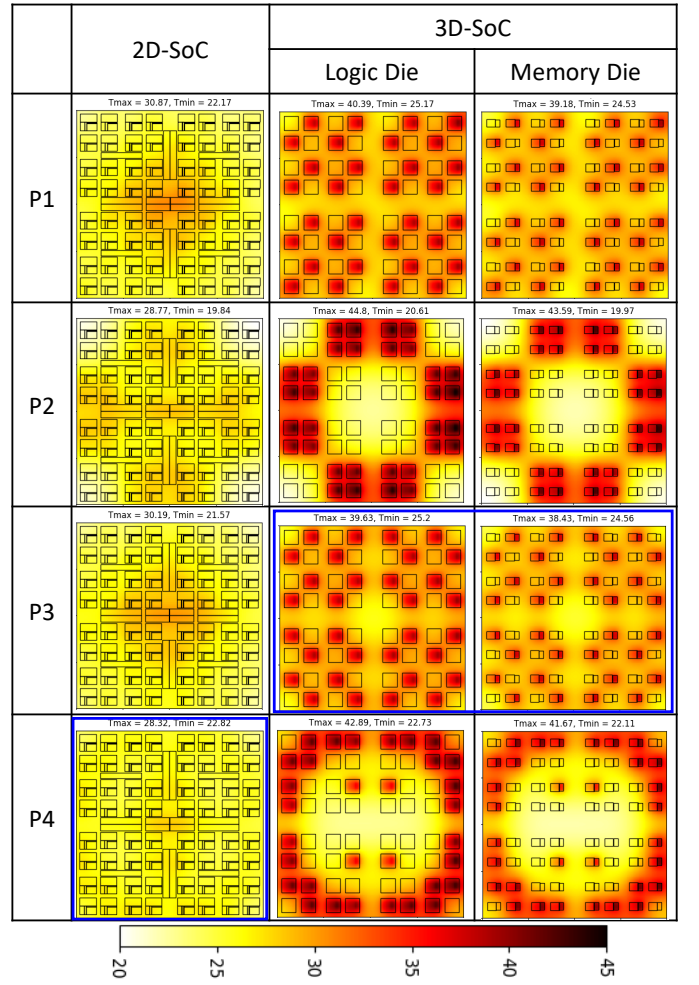
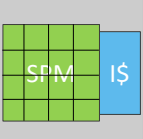
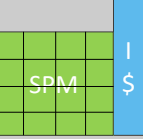


Figure 8: Temperature maps of the 2D and 3D-SOC for the proposed four different tile activity patterns under half (50%) workload.

2D-SOC. Also the different tile activity pattern results reveal that the pattern P4 for the 2D-SOC and pattern P3 for the 3D-SOC are exhibiting the optimum thermal performance as highlighted in Fig. 7. We have also tested the proposed tiles activity patterns by simulating the SOC with 25% workload conditions. The simulation results shown in Fig. 7(c) and 7(d) reveal that for 25% workload also the optimum tile pattern is similar to that in 50% workload. Also from Fig. 8, we observe a uniform on-chip temperature map for tile activity patterns P4 for 2D-SOC and pattern P3 for 3D-SOC compared to the other tile activity patterns. Therefore, these results strongly reveal that, although the overall combined workload is constant, the tiles/core activity pattern significantly impact the thermal performance in both 2D and 3D-SOC.

IV. L1 MEMORY CAPACITY ENHANCEMENT ON THERMAL PERFORMANCE OF 3D-SOC

For MemPool the shared-L1 memory capacity can be increased from 1 MiB to 8 MiB of SPM, which is reported in [41]. Here, the tiles are implemented to achieve a standard cell density of 90%. In this section we investigate the thermal performance by considering the impact of L1 SPM memory capacities of 1MiB, 2MiB, 4MiB and 8MiB.

	SPM Capacity		
	1, 2 MiB	4 MiB	8 MiB
Logic Die			
Memory Die			
Footprint*	0.667	0.767	0.933
Power*	0.91, 0.96	1.04	1.17

* Normalized with respect to MemPool 2D - base line

Figure 9: Floorplan of the memory die and logic die used to implement MemPool 3D-SOC along with the die footprint and power with L1 memory capacity of 1MiB, 2MiB, 4MiB, and 8 MiB.

A. MemPool-3D_{1,2MiB}

Fig. 9 shows the logic die and memory die floorplan for the MemPool-3D configuration with SPM capacity from 1MiB to 8 MiB. From the figure it is observed that the floorplan remains the same for 1MiB and 2MiB and the only difference is the utilization of the memory die, which is 51% and 65%, respectively. However, the power consumption in MemPool-3D_{1MiB} is still lower (0.91x) compared to MemPool-2D_{1MiB} due to a reduction of the number of buffers, owing to the decrease in interconnect wiring length. But from Fig. 10(a), the computed power density in MemPool-3D_{1MiB} is 37% higher than for 2D_{1MiB} due to a decrease in die area footprint. Therefore, the on-chip thermal performance in 3D_{1MiB} is expected to degrade as shown in Fig. 4. For MemPool-3D_{2MiB}, the increase in utilization of the memory die further increases the power consumption (i.e., from 0.91x to 0.96x) and results in a rise in memory die power density by 48%. Therefore, the increase in power density in the memory die and increased heat coupling from memory die to logic die leads to a further rise in on-chip T_{max} and ΔT in both logic and memory die compared to the 3D_{1MiB}.

B. MemPool-3D_{4MiB}

For an SPM capacity of 4MiB, the changes in the memory die floorplan achieve higher utilization (89%) with only 13% larger tile footprint than 1MiB and 2MiB. However, this increase in footprint reduces the power density for 3D_{4MiB} compared to 3D_{2MiB} but is still 62% larger compared to 2D_{4MiB} as shown in Fig. 10(a). It is also observed that although the power density for 3D_{4MiB} is smaller compared to 3D_{2MiB}, the on-chip temperatures (T_{max} and ΔT) are still larger for 3D_{4MiB}. This is due to the combined effect of the change in the floorplan of memory die and increase in tile utilization.

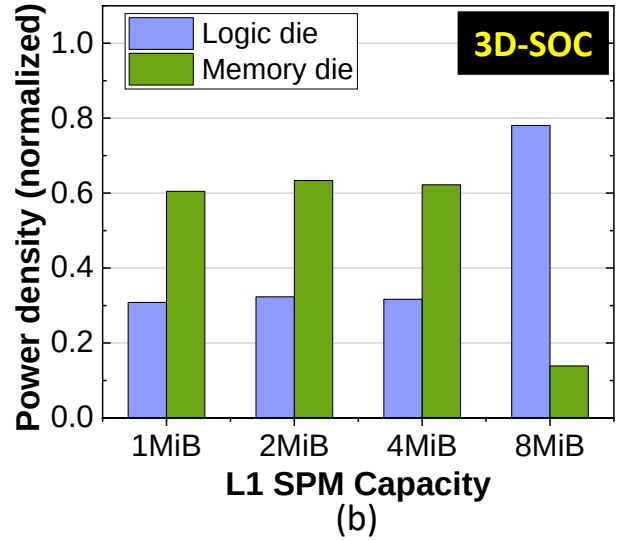
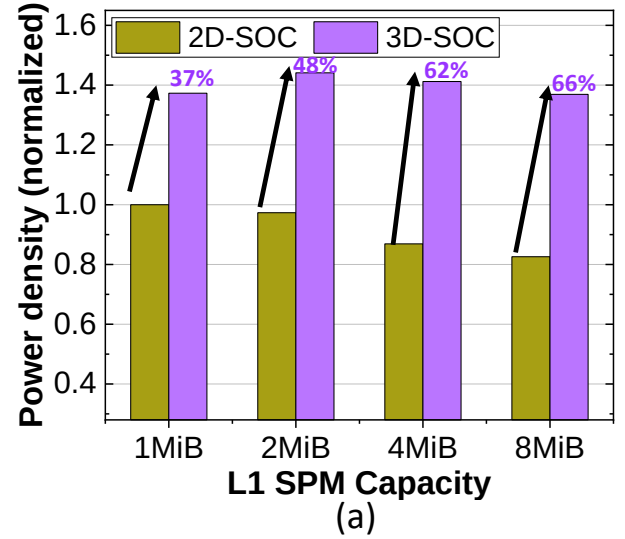


Figure 10: (a) The power density (normalized with respect to 2D_{1MiB}) versus the L1 SPM capacity enhancement from 1 MiB to 8 MiB for both MemPool 2D and 3D-SOC. (b) Logic die and memory die power density (normalized with respect to 2D_{1MiB}) of MemPool 3D-SOC for L1 SPM capacity enhancement from 1 MiB to 8 MiB. Data is extracted from the reference [41].

C. MemPool-3D_{8MiB}

However, for 8MiB case, the partitioning is completely different compared to previous cases. Here, due to the increase in memory macro size, the memory die is occupied by only 15 out of 16 SPM macros with 5×3 array with nearly 100% utilization. This also increases the die footprint by 40% compared to the MemPool-3D_{1MiB} which is very significant, especially when compared to the relatively small increase for the 4MB case.. The additional area in the logic die due to the rise in footprint area will be occupied by the remaining 1 SPM macro and all the tile's IS as shown in Fig. 9. Both the increase in power and footprint results in a decrease in power density for MemPool-3D_{8MiB} compared to 3D_{1MiB}, 3D_{2MiB} and 3D_{4MiB} but it is quite large (66%) compared to 2D_{8MiB}. From the HOTSPOT thermal simulation results shown in Fig. 11, there is a significant rise in on-chip T_{max}

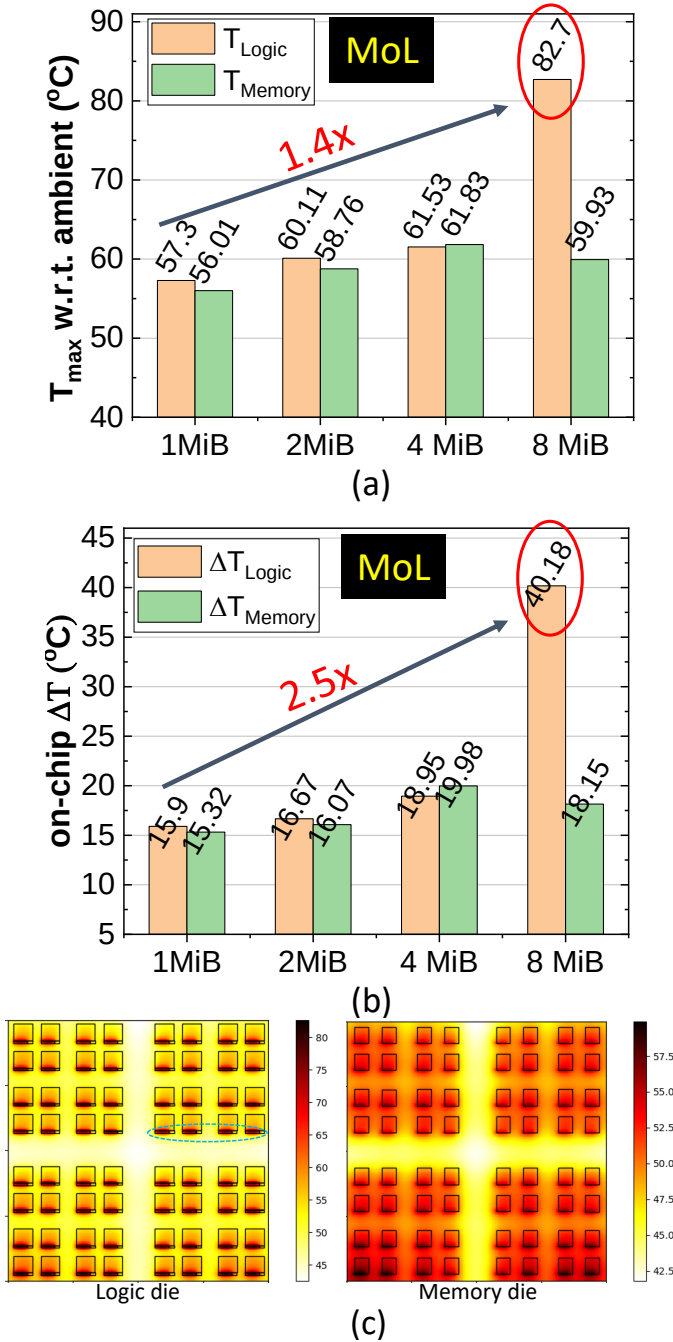


Figure 11: (a) The junction to ambient temperature (T_{max}) and (b) the on-chip thermal gradient (ΔT) versus the L1 memory capacity enhancement from 1 MiB to 8 MiB for the MemPool-3D SOC. (c) Logic die and memory die temperature map for MemPool-3D_{8MiB} highlighting that the logic die is hotter than the memory die. Note that the colorbar for logic die and memory die in (c) are different.

and thermal gradients and this is mainly due to the changed functional partitioning, in which the I\$ block is moved to the logic die and hence results in a larger power density in the logic die as shown in Fig. 10(b). The location of the hotspot is observed within the I\$ block of each tile in the memory die as shown in Fig. 11(c).

In summary, we observed that both the power consumption and footprint of the SOC scales with increasing the SPM L1

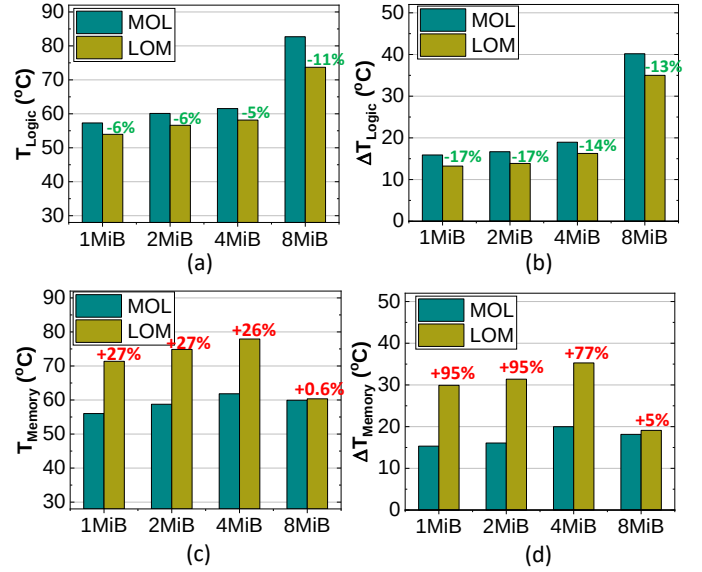


Figure 12: Thermal performance comparison between Memory-on-Logic (MOL) and Logic-on-Memory (LOM) by computing (a, b) the logic die temperatures, and (b, c) memory die temperatures for L1 SPM capacities of 1 MiB, 2 MiB, 4 MiB and 8 MiB. MOL shows improved thermal performance for 1 MiB, 2 MiB and 4 MiB whereas the LOM shows improved thermal performance compared to MOL.

memory capacity but not in linear way. The trend is quite complex actually as shown in Fig.11 (a)-(b). Especially, in the 2D-SOC reported in [41], the footprint increases faster than the power consumption which results in a decrease in power density as the memory capacity increases from 1 MiB to 8 MiB as shown in Fig. 10(a) and hence the on-chip T_{max} decreases as reported in in our previous study [38]. On the other hand, in the 3D-SOCs reported in this work, the on-chip thermal performance is determined not only by the power density but also by the 3D functional partitioning, especially in MemPool-3D_{8MiB} which is shown in Fig. 11.

V. IMPACT OF THE 3D-STACKING CONFIGURATIONS

So far in the above 3D stacking configuration, the logic die is placed at the bottom and the memory die is on the top side (i.e., MOL configuration). However, with this configuration the logic die is at a farther distance from the cooling solution resulting in a higher thermal resistance for the logic die. Also, the Si thinning in the logic layer further degrades the lateral heat spreading. All these effects significantly impact the system thermal performances in the 3D-SOC. Especially in MemPool-3D_{8MiB}, the power dissipation in the logic die is also substantially larger than that of in the Memory die due to the presence of the power-hungry I\$ block in the logic die (Fig. 10(b)). Therefore, one possible solution to reduce the maximum temperature is, by placing the power-hungry logic die on top of the memory die (i.e., LOM), which is closer to the cooling solution; also the absence of Si thinning may improve the overall system thermal performance [52]. However, a Si thinning is now necessary in the memory die in order to communicate to the bottom IO layer through the nano-TSVs.

Fig. 12 shows the thermal performance comparison between MOL and LOM for L1 Memory capacity of from 1 MiB, 2 MiB, 4 MiB and 8 MiB. From the figure, for LOM configuration, the logic die thermal performance is improved by decreasing the die temperatures (i.e., T_{Logic} and ΔT_{Logic}) for all the SPM capacities (Fig. 12(a) and Fig. 12(b)). Whereas, the memory die performance is degraded greatly by increasing the die temperatures (i.e., $T_{Memory} > 25\%$ and $\Delta T_{Memory} > 75\%$) for SPM capacities 1MiB, 2MiB and 4MiB (Fig. 12(c) and Fig. 12(d)). However, for SPM L1 capacity of 8MiB, the thermal performance is improved by reducing the logic die temperature (T_{Logic}) by 11% with insignificant rise in memory die temperature (i.e., $T_{Logic} < 1\%$). Therefore, the simulation results reveals that the thermal issues can be mitigated in the proposed MemPool 3D-SOC with MOL configuration for SPM L1 capacity of 1MiB, 2MiB and 4MiB, and with LOM configuration for SPM L1 capacity of 8MiB. These results are good agreement with the literature data [31].

VI. CONCLUSION

We have performed a detailed thermal modeling study for both 2D and 3D SOC with MemPool as a test case and compared the results between 2D-SOC and 3D-SOC. The simulation results reveal that several thermal issues are predominant in 3D SOC. In MemPool-3D SOC with memory on logic configuration the maximum system temperature is increased by more than 20%. It is also observed that the logic die is hotter than the memory die due to the presence of higher power and farther distance from the cooling solution. Also, the poor lateral heat spreading due to silicon thinning in the bottom logic layer is also an additional impact for poor thermal performance. Investigation of the tile-level activity patterns on thermal performance on both 2D and 3D-SOC reveals that pattern P4 for 2D-SOC and pattern 3 for 3D-SOC are optimum among the four patterns considered in this work under 50% workload conditions. The architectural floor-planning impact on the thermal performance with different memory capacities ranging 1 MiB to 8 MiB reveals that the the system maximum temperature is increased by 44% and it is predominant for SPM capacity of 8MiB. Finally, the impact of 3D-SOC stacking configurations such as MOL and LOM on thermal performances reveal that the MOL configuration showed an improved thermal performance for SPM L1 capacity of 1 MiB, 2 MiB and 4 MiB, whereas for SPM L1 capacity of 8 MiB, the LOM the system thermal performance is improved by more than 10% compared to the MOL configuration.

ACKNOWLEDGMENTS

This work of Jun-Han Han and Mircea R. Stan was supported in part by the Semiconductor Research Corporation (SRC) Joint University Microelectronics Program (JUMP) Task 2780.003: Thermal and Power Management.

REFERENCES

- [1] A. Acovic, G. La Rosa, and Y.-C. Sun, "A review of hot-carrier degradation mechanisms in MOSFETs," *Microelectronics Reliability*, vol. 36, no. 7-8, pp. 845–869, 1996, doi: [https://doi.org/10.1016/0026-2714\(96\)00022-4](https://doi.org/10.1016/0026-2714(96)00022-4).
- [2] K. F. Schuegraf and C. Hu, "Reliability of thin SiO₂," *Semiconductor Science and Technology*, vol. 9, no. 5, p. 989, 1994, doi: [10.1088/0268-1242/9/5/002](https://doi.org/10.1088/0268-1242/9/5/002).
- [3] T. Grasser, B. Kaczer, W. Goes, H. Reisinger, T. Aichinger, P. Hehenberger, P.-J. Wagner, F. Schanovsky, J. Franco, M. T. T. Luque *et al.*, "The paradigm shift in understanding the bias temperature instability: From reaction-diffusion to switching oxide traps," *IEEE Transactions on Electron Devices*, vol. 58, no. 11, pp. 3652–3666, 2011, doi: [10.1109/TED.2011.2164543](https://doi.org/10.1109/TED.2011.2164543).
- [4] S. Venkateswarlu and K. Nayak, "Device SHEs in the Presence of Non-equilibrium Channel Heat Transport in SOI and SOD FinFETs with Technology Scaling," in *2020 5th IEEE International Conference on Emerging Electronics (ICEE)*. IEEE, 2020, pp. 1–4, doi: [10.1109/ICEE50728.2020.9777080](https://doi.org/10.1109/ICEE50728.2020.9777080).
- [5] —, "Ambient temperature-induced device self-heating effects on multi-fin Si CMOS logic circuit performance in N-14 to N-7 scaled technologies," *IEEE Transactions on Electron Devices*, vol. 67, no. 4, pp. 1530–1536, 2020, doi: [10.1109/TED.2020.2975416](https://doi.org/10.1109/TED.2020.2975416).
- [6] H. Esmaeilzadeh, E. Blem, R. S. Amant, K. Sankaralingam, and D. Burger, "Dark silicon and the end of multicore scaling," *IEEE Micro*, vol. 32, no. 3, pp. 122–134, 2012, doi: <https://doi.org/10.1145/2000064.2000108>.
- [7] S. Borkar, "Thousand core chips: a technology perspective," in *Proceedings of the 44th annual design automation conference*, 2007, pp. 746–749, doi: <https://doi.org/10.1145/1278480.1278667>.
- [8] E. Beyne, "The 3-D interconnect technology landscape," *IEEE Design & Test*, vol. 33, no. 3, pp. 8–20, 2016, doi: [10.1109/MDAT.2016.2544837](https://doi.org/10.1109/MDAT.2016.2544837).
- [9] D. Zhang and J. J.-Q. Lu, "3D integration technologies: An overview," *Materials for Advanced Packaging*, pp. 1–26, 2017, doi: https://doi.org/10.1007/978-3-319-45098-8_1.
- [10] V. F. Pavlidis and E. G. Friedman, "Interconnect-based design methodologies for three-dimensional integrated circuits," *Proceedings of the IEEE*, vol. 97, no. 1, pp. 123–140, 2009, doi: [10.1109/JPROC.2008.2007473](https://doi.org/10.1109/JPROC.2008.2007473).
- [11] A. Rahman and R. Reif, "Thermal analysis of three-dimensional (3-D) integrated circuits (ICs)," in *Proceedings of the IEEE 2001 International Interconnect Technology Conference (Cat. No. 01EX461)*. IEEE, 2001, pp. 157–159, doi: [10.1109/IITC.2001.380045](https://doi.org/10.1109/IITC.2001.380045).
- [12] P. Morrow, C.-M. Park, S. Ramanathan, M. Kobrinsky, and M. Harmes, "Three-dimensional wafer stacking via Cu-Cu bonding integrated with 65-nm strained-Si/low-k CMOS technology," *IEEE electron device letters*, vol. 27, no. 5, pp. 335–337, 2006, doi: [10.1109/LED.2006.873424](https://doi.org/10.1109/LED.2006.873424).
- [13] S. Pozder, R. Chatterjee, A. Jain, Z. Huang, R. E. Jones, and E. Acosta, "Progress of 3D integration technologies and 3D interconnects," in *2007 IEEE International Interconnect Technology Conference*. IEEE, 2007, pp. 213–215, doi: [10.1109/IITC.2007.382393](https://doi.org/10.1109/IITC.2007.382393).
- [14] K. Guanari *et al.*, "Electrical integrity of state-of-the-art 0.13 um SOI CMOS devices and circuits transferred for three-dimensional (3D) integrated circuit (IC) fabrication," in *Proc. Int. Electron Devices Meeting, IEDM*, 2002, pp. 943–945, doi: [10.1109/IEDM.2002.1175992](https://doi.org/10.1109/IEDM.2002.1175992).
- [15] V. Venkatadri, B. Sammakia, K. Srihari, and D. Santos, "A review of recent advances in thermal management in three dimensional chip stacks in electronic systems," 2011, doi: <https://doi.org/10.1115/1.4005298>.
- [16] S. G. Kandlikar, "Review and projections of integrated cooling systems for three-dimensional integrated circuits," *Journal of Electronic Packaging*, vol. 136, no. 2, p. 024001, 2014, doi: <https://doi.org/10.1115/1.4027175>.
- [17] S. Jeloka, B. Cline, S. Das, B. Labbe, A. Rico, R. Herberholz, J. DeLaCruz, R. Mathur, and S. Hung, "System technology co-optimization and design challenges for 3D IC," in *2022 IEEE Custom Integrated Circuits Conference (CICC)*. IEEE, 2022, pp. 1–6, doi: [10.1109/CICC53496.2022.9772831](https://doi.org/10.1109/CICC53496.2022.9772831).
- [18] S. Kochupurackal Rajan, "DENSE INTERCONNECTION AND ADVANCED COOLING TECHNOLOGIES FOR 2.5 D AND 3D HETEROGENEOUS INTEGRATED CIRCUITS," Ph.D. dissertation, Georgia Institute of Technology, 2022, doi: <http://hdl.handle.net/1853/66653>.
- [19] D. Brooks and M. Martonosi, "Dynamic thermal management for high-performance microprocessors," in *Proceedings HPCA Seventh International Symposium on High-Performance Computer Architecture*. IEEE, 2001, pp. 171–182, doi: [10.1109/HPCA.2001.903261](https://doi.org/10.1109/HPCA.2001.903261).
- [20] K. Skadron, M. R. Stan, K. Sankaranarayanan, W. Huang, S. Velusamy, and D. Tarjan, "Temperature-aware microarchitecture: Modeling and implementation," *ACM Transactions on Architecture and Code Optimization (TACO)*, vol. 1, no. 1, pp. 94–125, 2004, doi: <https://doi.org/10.1145/980152.980157>.
- [21] J. Donald and M. Martonosi, "Techniques for multicore thermal management: Classification and new exploration," *ACM SIGARCH computer*

- architecture news*, vol. 34, no. 2, pp. 78–88, 2006, doi: 10.1109/ISCA.2006.39.
- [22] G. M. Link and N. Vijaykrishnan, “Thermal trends in emerging technologies,” in *7th International Symposium on Quality Electronic Design (ISQED’06)*. IEEE, 2006, pp. 8–pp, doi: 10.1109/ISQED.2006.136.
 - [23] C. Zhu, Z. Gu, L. Shang, R. P. Dick, and R. Joseph, “Three-dimensional chip-multiprocessor run-time thermal management,” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 27, no. 8, pp. 1479–1492, 2008, doi: 10.1109/TCAD.2008.925793.
 - [24] K. Puttaswamy and G. H. Loh, “Thermal herding: Microarchitecture techniques for controlling hotspots in high-performance 3D-integrated processors,” in *2007 IEEE 13th International Symposium on High Performance Computer Architecture*. IEEE, 2007, pp. 193–204, doi: 10.1109/HPCA.2007.346197.
 - [25] A. K. Coskun, J. L. Ayala, D. Atienza, T. S. Rosing, and Y. Leblebici, “Dynamic thermal management in 3D multicore architectures,” in *2009 Design, Automation & Test in Europe Conference & Exhibition*. IEEE, 2009, pp. 1410–1415, doi: 10.1109/DATE.2011.5763053.
 - [26] F. Hameed, M. A. Al Faruque, and J. Henkel, “Dynamic thermal management in 3D multi-core architecture through run-time adaptation,” in *2011 Design, Automation & Test in Europe*. IEEE, 2011, pp. 1–6, doi: 10.1109/DATE.2011.5763053.
 - [27] D. H. Kim, S. Kim, and S. K. Lim, “Impact of nano-scale through-silicon vias on the quality of today and future 3D IC designs,” in *International Workshop on System Level Interconnect Prediction*. IEEE, 2011, pp. 1–8, doi: 10.1109/SLIP.2011.6135435.
 - [28] M. Murugesan, K. Mori, T. Kojima, H. Hashimoto, J. Bea, T. Fukushima, and M. Koyanagi, “Nano Ni/Cu-TSVs with an improved reliability for 3D-IC integration application,” in *2020 31st Annual SEMI Advanced Semiconductor Manufacturing Conference (ASMC)*. IEEE, 2020, pp. 1–5, doi: 10.1109/ASMC49169.2020.9185397.
 - [29] G. Sisto, R. Chen, R. Chou, G. Van der Plas, E. Beyne, R. Metcalfe, and D. Milojevic, “Design And Sign-off Methodologies For Wafer-To-Wafer Bonded 3D-ICs At Advanced Nodes,” in *2021 ACM/IEEE International Workshop on System Level Interconnect Prediction (SLIP)*. IEEE, 2021, pp. 17–23, doi: 10.1109/SLIP52707.2021.00011.
 - [30] A. Jain, R. E. Jones, R. Chatterjee, S. Pozder, and Z. Huang, “Thermal modeling and design of 3D integrated circuits,” in *2008 11th Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems*. IEEE, 2008, pp. 1139–1145, doi: 10.1109/ITHERM.2008.4544389.
 - [31] R. Mathur, C.-J. Chao, R. Liu, N. Tadepalli, P. Chandupatla, S. Hung, X. Xu, S. Sinha, and J. Kulkarni, “Thermal analysis of a 3D stacked high-performance commercial microprocessor using face-to-face wafer bonding technology,” in *2020 IEEE 70th Electronic Components and Technology Conference (ECTC)*. IEEE, 2020, pp. 541–547, doi: 10.1109/ECTC32862.2020.00091.
 - [32] L. Zhu, T. Ta, R. Liu, R. Mathur, X. Xu, S. Das, A. Kaul, A. Rico, D. Joseph, B. Cline *et al.*, “Power delivery and thermal-aware arm-based multi-tier 3D architecture,” in *2021 IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*. IEEE, 2021, pp. 1–6, doi: 10.1109/ISLPED52811.2021.9502481.
 - [33] Y. Cho, K. Lee, K. Hwang, H. Lee, Y. Im, and M. Kim, “Thermal perspective design Analysis of 3D stacked chip integrated by TSV, Micro-bump and TMV with SOC,” in *2022 21st IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm)*. IEEE, 2022, pp. 1–6, doi: 10.1109/iTherm54085.2022.9899663.
 - [34] J.-H. Han, X. Guo, K. Skadron, and M. R. Stan, “From 2.5 D to 3D Chiplet Systems: Investigation of Thermal Implications with HotSpot 7.0,” in *2022 21st IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm)*. IEEE, 2022, pp. 1–6, doi: 10.1109/iTherm54085.2022.9899649.
 - [35] P. Bogdan, “Mathematical modeling and control of multifractal workloads for data-center-on-a-chip optimization,” in *Proceedings of the 9th International Symposium on Networks-on-Chip*, 2015, pp. 1–8, doi: https://doi.org/10.1145/2786572.2786592.
 - [36] S. Corbetta and W. Fornaciari, “Estimation of thermal status in multi-core systems,” in *2011 IEEE International Symposium of Circuits and Systems (ISCAS)*. IEEE, 2011, pp. 1660–1663, doi: 10.1109/ISCAS.2011.5937899.
 - [37] M.-H. Haghighyan, A. Miele, Z. Zou, H. Tenhunen, and J. Plosila, “Thermal-cycling-aware dynamic reliability management in many-core system-on-chip,” in *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2020, pp. 1229–1234, doi: 10.23919/DATE48585.2020.9116325.
 - [38] S. Venkateswarlu, S. Mishra, H. Oprins, B. Vermeersch, M. Brunion, J.-H. Han, M. R. Stan, P. Weckx, and F. Catthoor, “Thermal Performance Analysis of Mempoool RISC-V Multicore SoC,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 30, no. 11, pp. 1668–1676, 2022, doi: 10.1109/TVLSI.2022.3207553.
 - [39] S. H. Pan, N. Chang, and T. Hitomi, “3D-IC dynamic thermal analysis with hierarchical and configurable chip thermal model,” in *2013 IEEE International 3D Systems Integration Conference (3DIC)*. IEEE, 2013, pp. 1–8, doi: 10.1109/3DIC.2013.6702373.
 - [40] M. Cavalcante, S. Riedel, A. Pullini, and L. Benini, “MemPool: A Shared-L1 Memory Many-Core Cluster with a Low-Latency Interconnect,” in *2021 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2021, pp. 701–706, doi: 10.23919/DATE51398.2021.9474087.
 - [41] M. Cavalcante, A. Agnesina, S. Riedel, M. Brunion, A. Garcia-Ortiz, D. Milojevic, F. Catthoor, S. K. Lim, and L. Benini, “MemPool-3D: Boosting Performance and Efficiency of Shared-L1 Memory Many-Core Clusters with 3D Integration,” *arXiv preprint arXiv:2112.01168*, 2021, doi: https://doi.org/10.48550/arXiv.2112.01168.
 - [42] L. Bamberg, A. Garcia-Ortiz, L. Zhu, S. Pentapati, S. K. Lim *et al.*, “Macro-3D: A physical design methodology for face-to-face-stacked heterogeneous 3D ICs,” in *2020 Design, Automation & Test in Europe Conference & Exhibition (DATE)*. IEEE, 2020, pp. 37–42, doi: 10.23919/DATE48585.2020.9116297.
 - [43] B. Vermeersch, J. Carrete, and N. Mingo, “Cross-plane heat conduction in thin films with ab-initio phonon dispersions and scattering rates,” *Applied Physics Letters*, vol. 108, no. 19, p. 193104, 2016, doi: https://doi.org/10.1063/1.4948968.
 - [44] K. W. Yan, P.-Y. Lin, and S.-L. Kuo, “Thermal Challenges for HPC 3DFabric TM Packages and Systems,” in *2022 IEEE International Reliability Physics Symposium (IRPS)*. IEEE, 2022, pp. 4C–1, doi: 10.1109/IRPS48227.2022.9764572.
 - [45] “HOTSPOT,” <https://github.com/uva-hotspot/HotSpot>.
 - [46] J.-H. Han, R. E. West, K. Skadron, and M. R. Stan, “Thermal Simulation of Processing-in-Memory Devices using HotSpot 7.0,” in *2021 27th International Workshop on Thermal Investigations of ICs and Systems (THERMINIC)*. IEEE, 2021, pp. 1–5, doi: 10.1109/THERMINIC52472.2021.9626520.
 - [47] K. Skadron, K. Sankaranarayanan, M. Stan, R. Ribando, and W. Huang, “An improved hotspot block-based thermal model with granularity considerations (extended discussion),” 2007.
 - [48] V. Székely, C. Márta, M. Rencz, G. Végh, Z. Benedek, and S. Torok, “A thermal benchmark chip: Design and applications,” *IEEE Transactions on Components, Packaging, and Manufacturing Technology: Part A*, vol. 21, no. 3, pp. 399–405, 1998, doi: 10.1109/95.725202.
 - [49] H. Oprins, D. Milojevic, G. Van der Plas, and E. Beyne, “Thermal analysis of 3D functional partitioning for high-performance systems,” in *2021 20th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (iTherm)*. IEEE, 2021, pp. 145–153, doi: 10.1109/iTherm51669.2021.9503209.
 - [50] W. L. Bircher and L. John, “Predictive power management for multi-core processors,” in *International Symposium on Computer Architecture*. Springer, 2010, pp. 243–255, doi: https://doi.org/10.1007/978-3-642-24322-6_21.
 - [51] W. L. Bircher and L. K. John, “Core-level activity prediction for multicore power management,” *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, vol. 1, no. 3, pp. 218–227, 2011, doi: 10.1109/JETCAS.2011.2164973.
 - [52] L. Zhu, L. Bamberg, S. S. K. Pentapati, K. Chang, F. Catthoor, D. Milojevic, M. Komalan, B. Cline, S. Sinha, X. Xu *et al.*, “High-Performance Logic-on-Memory Monolithic 3-D IC Designs for Arm Cortex-A Processors,” *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 29, no. 6, pp. 1152–1163, 2021, doi: 10.1109/TVLSI.2021.3073070.